

Multimodal Situational Awareness: Neuro-Symbolic AI for Real-Time HCI in the Classroom

Yifan Zhu¹[0009–0000–6597–7275], Kenneth Lai¹[0000–0003–2870–7019], Ibrahim
Khebour²[0009–0009–4374–7263], Marc Verhagen¹[0000–0002–2284–8163], Nikhil
Krishnaswamy,²[0000–0001–7878–7227], and James
Pustejovsky¹[0000–0003–2233–9761]

¹ Brandeis University, Waltham, MA 02453, USA

² Colorado State University, Fort Collins, CO 80523, USA
{zhuyifan,jamesp}@brandeis.edu

Abstract. This research addresses the problem of an enabling AI systems to become capable of dynamically adapting to the complex, multimodal nature of classroom interactions. We introduce a novel neuro-symbolic AI approach designed to achieve real-time, multi-party situational awareness in educational environments. By integrating verbal, gestural, and physical cues from multiple students, our system constructs a coherent understanding of collaborative learning processes. This involves not only tracking verbal dialogue but also interpreting nonverbal behavior and contextual changes to analyze interaction dynamics. The proposed framework advances the development of AI systems that can effectively support and enhance classroom engagement and learning outcomes.

Keywords: Situational awareness · Multimodality · AI in education · Non-verbal behavior · Gesture recognition · Gaze detection

1 Introduction

Multimodal methods have gained increasing prominence in machine learning-based situational awareness (SA) systems, significantly enhancing capabilities in perception, comprehension, and prediction [7–9, 15, 23, 24]. By integrating diverse data sources, these approaches offer a more comprehensive representation of the context. Multimodal fusion methods have been shown to substantially improve the scope and reliability of predictive outcomes in SA tasks [13, 21, 23]. Furthermore, these techniques have demonstrated notable advantages in enhancing model interpretability and explainability [6, 14], solidifying their critical role in advancing SA-related research and applications.

While artificial intelligence (AI)-based multimodal techniques have been extensively employed in domains such as military operations, healthcare, and industrial monitoring [16, 19, 33], their application within the field of education remains comparatively underexplored. Notably, AI-driven multimodal approaches

have not yet been widely adopted for the comprehensive analysis of engagement levels in classroom interactions. Conventional methods predominantly depend on behavioral observations and self-reported data, which are often prone to subjectivity and inconsistency [10, 11].

Developing a well-defined model over multiple modalities presents significant challenges due to the inherent complexities of each individual modality. Sparse and insufficient data for alignment across modalities is a primary issue. For example, in multi-party classroom settings, the audio speech component poses difficulties for current automatic speech recognition (ASR) models for two key reasons: (1) these models are typically not trained on multi-speaker data, particularly data from classroom environments with students aged 10–15 [34], and (2) a significant portion of the data we aim to leverage consists of non-verbal behaviors, which are not adequately represented in existing datasets. While large public datasets exist for recognizing non-verbal behaviors in specific crowd settings, these datasets are not tailored to classroom environments or behaviors indicative of engagement. Moreover, these datasets often include only two modalities (e.g., speech and gesture), whereas classroom engagement analysis requires additional modalities such as posture, gaze, and action, for which there is currently no available data. These gaps highlight the need for a robust method to integrate multiple modalities effectively.

One promising approach is the use of neuro-symbolic mechanisms. Neural models can be employed to recognize individual components from each modality, such as speech, action, posture, gesture, and gaze. Beyond this, we argue that the interplay between robust neural techniques and inherent symbolic reasoning provides a more effective framework for modeling higher-order epistemic states, such as beliefs and intentions, which are critical for understanding interactions in classroom settings. This integration not only addresses the challenges of data sparsity and alignment but also enhances the model’s ability to reason about complex social and cognitive dynamics in real-world scenarios.

To address the pressing need for AI systems capable of dynamically adapting to the complex, multimodal nature of classroom interactions [3], we introduce a novel neuro-symbolic AI approach designed to achieve real-time, multi-party situational awareness in educational environments. By integrating verbal, gestural, and physical cues from multiple students [12, 17], our system constructs a coherent understanding of collaborative learning processes. This involves not only tracking verbal dialogue but also interpreting nonverbal signals and contextual changes to analyze interaction dynamics [18, 28, 29].

The proposed framework enables the AI to capture the evolving situational context and the group’s shared mental model [30, 38]. By employing neuro-symbolic AI, the system combines the perceptual power of neural networks with the reasoning capabilities of symbolic AI. This dual approach allows the AI to reason about team dynamics while grounding its understanding in the physical environment, such as interpreting deictic references.

2 Related Work

Previous studies have explored various multimodal approaches for situational awareness. For instance, posture detection and classification in smart home environments have been introduced in [4]. Eye-tracking signals combined with electroencephalogram (EEG) data have been employed to assess workload perception in air traffic control officers [20]. Furthermore, a novel cancelable biometric template generation algorithm integrating random biometric fusion, random projection, and selection has been developed for SA applications in both real and virtual environments [27].

In the context of human-computer interaction, multimodal approaches have shown significant potential for improving situational awareness. For instance, [13] highlights the benefits of integrating multiple modalities, such as speech, environmental audio, and handwriting input, to enhance SA capabilities. Similarly, [37] proposes a multimodal sensing framework designed for objective monitoring of situational awareness. Further advancing this field, [36] introduces an attention-enabled recognition network with a decision-level fusion architecture to improve workload estimation performance. Additionally, time-step level fusion has been applied to generate fused features for sentiment classification at the utterance level [23]. Audio and video signals, along with biopsychological data, have also been utilized to detect intra-individual behavioral prototypes for predicting users' emotional states [6]. These applications underscore the value of multimodal methods in advancing HCI research and practice.

While multimodal techniques have shown promise in various domains, their application to evaluating engagement levels in classroom settings remains limited. Current approaches to assessing student engagement often rely on traditional methods, such as behavioral observations, self-reported surveys, and manual coding of interactions, which are informed by conceptual frameworks that define engagement and disengagement in terms of students' activation and pleasure [1]. These methods, while useful, are often subjective, time-consuming, and unable to capture the dynamic and multifaceted nature of engagement in real time.

3 Neuro-symbolic network (NSN) architecture

Our proposed neuro-symbolic network (NSN) architecture (see Fig. 1) is a framework that synergistically combines neural networks' pattern recognition capabilities with symbolic reasoning's explicit knowledge representations. In the context of multimodal multiparty dialogues involving tracking common ground, epistemics, object tracking, co-attention (gaze), and speech utterances, we argue that an NSN processes and integrates information across different modalities to model and reason about complex social interactions more effectively than current neural network architectures. The NSN architecture can be divided into three primary components: the multimodal feature extractor, the multimodal dense paraphrasing module, and the common ground tracking mechanism.

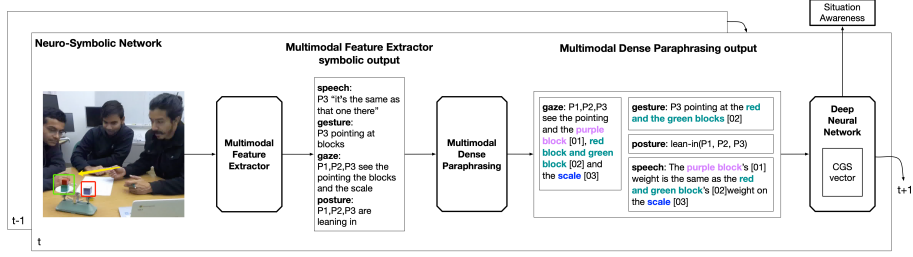


Fig. 1: Neuro-symbolic network (NSN) architecture.

3.1 Multimodal Feature Extractor

The proposed feature extractor is designed to extract multimodal features from existing visual and auditory data. The pipeline incorporates two complementary methods for information extraction: modality-specific extractors tailored to process distinct features from individual modalities, such as gaze, posture, gestures, and speech; and manual annotation for actions.

The processing mechanism for each modality is detailed below:

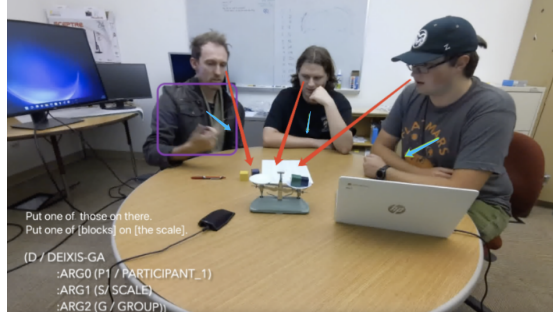


Fig. 2: Example of a multimodal annotated situation. The red arrows represent gaze, the blue arrows, posture (leaning in), and the purple box, gesture.

Speech processing consists of two primary steps. First, speech diarization with Pyannote [5] is conducted to distinguish and segment audio data by individual speakers. Second, speech recognition using Whisper [31] is applied to transcribe the segmented audio into textual form. This two-step process enables the extraction of meaningful linguistic information from multi-speaker audio inputs.

Posture classification begins by leveraging the body joint data extracted using the Azure Kinect SDK. We segment the participants' positions into three regions—left, middle, and right—based on the x-coordinate of each individual's pelvis. Following this, the positional and orientation data for each participant is flattened into feature vectors. These vectors are subsequently stacked and fed into a two-layer feedforward neural network. To account for variations across different positions, three separate models are trained, one for each participant position (left, middle, and right).

Gaze detection utilizes the direction of participants’ noses as a proxy for gaze estimation. Using the Azure Kinect SDK again, the average position of the two ear joints is calculated, resulting in a point approximately located behind the nose. A vector is then constructed between this point and the nose joint to approximate the gaze direction. This vector is extended into three-dimensional (3D) space to determine the objects or areas on which the participants’ gazes are focused.

Gesture recognition is done in two steps, starting from the detected joint positions. First, gesture detection identifies periods in the video with high rates of hand motion: these are the periods when gestures are likely to be occurring. Then, we perform gesture classification. We use MediaPipe [22] to detect additional, more fine-grained keypoints, e.g., knuckle positions on a hand. We pass these keypoints into a convolutional neural network to classify the gesture type. For pointing gestures, we additionally construct a pointing vector, similar to the gaze vector, to identify the object of the pointing.

Action annotation employs a tool designed to capture, categorize, and label human actions from closed set of predicate selected based on task requirements. This tool provides comprehensive temporal markers and unique identifiers, enabling efficient synchronization and analysis of action data alongside other modalities.

These symbolic outputs are then passed to the multimodal dense paraphrasing module, where the modalities are aligned and further contextualized to enrich the extracted information.

Using the scenario depicted in Fig. 1 as an example, the three participants in the video are first differentiated through speaker diarization. Subsequently, we recognize the following modalities:

- Speech: P3: It’s the same as that one there.
- Gaze: gaze(P1 and P2 and P3, scale)
- Gesture: point(P3, left_scale)
- Posture: lean-in(P1 and P2 and P3)

3.2 Multimodal Dense Paraphrasing (MMDP) Module

This module represents a key innovation within the NSN architecture. It incorporates cross-modal integration, which combines neural representations from different modalities to generate an aligned and enriched data structure from multimodal inputs. This process integrates and contextualizes information across modalities, as demonstrated in prior work [32, 35]. The fusion functions used for this integration can take various forms, including concatenation (combining features directly), attention mechanisms (dynamically weighting inputs based on their relevance), or learned transformations (using neural networks to learn optimal combinations).

After receiving symbolic representations from the multimodal feature extractor, the module first decontextualizes pronouns referring to task-relevant entities

in the dialogue through coreferential redescription. It then links the same entities across different modalities, providing alignment within the unimodal system. Subsequently, the model paraphrases mentions that refer to the same entity into their most informative form. This process aligns actions and gestures with the corresponding utterances that overlap most in terms of start and end times.

The mechanism incorporates two distinct forms of linguistic paraphrasing: **Machine Readable Paraphrase (MRP)** and **Human Readable Paraphrase (HRP)**. The MRP is a (semi-)structured textual representation of the multimodality expressed in the dialogue. It is generated as a set of key-value pairs that map each agent and modality to the content of the communicative event (e.g., action, utterance, gesture, etc.). The MRP features a uniform structure and text patterns that efficiently encode the semantics of multimodal interactions in a dialogue. This structured representation ensures consistency and facilitates computational processing. The MRP is subsequently converted into an HRP using Large Language Models (LLMs). The HRP encodes implicit semantics, leveraging LLMs’ capabilities to reconstruct sentence structures from often incomplete and disfluent speech, as well as to resolve anaphoric references across different modalities. The HRP, in its natural language form, is more effectively processed and interpreted by language models, enabling richer contextual understanding. These two forms of linguistic paraphrasing enable the alignment of information across different modalities. The symbolic output is enriched with contextualized information from all modalities and subsequently integrated into a dynamic common ground structure. This structure evolves at each timestamp, allowing the system to predict situational awareness and maintain a shared understanding across modalities.

Using the example in Fig. 1, the symbolic representations extracted from the modalities will be enriched through the dense paraphrasing mechanism in its MRP form as follows:

- Speech: P3: It[The purple block’s weight]’s the same as ~~that one~~ [the red and green blocks’ weight] ~~there~~[on the right scale].
- Gaze: gaze(P1 and P2 and P3, ~~scale~~[the purple block, red block, green block, scale, and pointing gesture])
- Gesture: point(P3, ~~left_scale~~[the red block and green block])
- Posture: lean-in(P1 and P2 and P3)

And the HRP form will be:

- Speech: Participant 3 concluded that the purple block’s weight’s the same as the red and green blocks’ weight.
- Gaze: Participant 1 and Participant 2 and Participant 3 all saw the purple block, the red block, the green block, the scale, and the pointing gesture.
- Gesture: Participant 3 points at the left scale, the red block and green block.
- Posture: Participant 1 and Participant 2 and Participant 3 all lean in.

3.3 Common Ground Tracking Module

The Common Ground (CG) Tracking module is a critical component in collaborative learning environments, where maintaining a shared understanding of tasks, instructions, and goals among participants is essential. We define the CG as:

$$CG = \bigcap_{a \in A} B_a,$$

where A is the set of all agents and B_a is the set of propositions that agent a believes.

To capture temporal dependencies in interactions, the module employs sequential models for temporal dynamics and dialogue modeling. These models are specifically designed to process sequential data and extract meaningful patterns over time, enabling the system to understand and predict the evolution of interactions.

Additionally, the module dynamically updates knowledge and beliefs as new information arrives. This ensures that the system remains adaptive and responsive to evolving interactions, allowing it to maintain an accurate and up-to-date representation of the context and participants' states.

This module first processes the enriched representations from the MMDP module as input and organizes the information into a Common Ground Structure (CGS) composed of three distinct banks: the **Evidence Bank (E-Bank)** stores all evidence derived from the enriched multimodal information; the **Fact Bank (F-Bank)** maintains updated shared beliefs and facts agreed upon during group interactions; and the **Question Bank (Q-Bank)** contains questions related to predefined task requirements, facilitating task-oriented collaboration [17].

This is a simplified version of evidence-based Dynamic Epistemic Logic [2, 26]. The Q-Bank is initialized with a set of predefined, task-relevant propositions, which are dynamically removed as the dialogue progresses and evidence for or against the propositions is introduced. A public announcement of a proposition comprises evidence for that proposition, and so moves the proposition from the Q-Bank to the E-Bank. Acceptance of that proposition, where acceptance can be verbal or implicit, then moves it from the E-Bank to the F-Bank [38].

The propositional representation of the epistemic state derived from the CGS is then passed to a recurrent deep neural network (DNN). The DNN dynamically tracks the evolution of the common ground over time, generating a symbolic output at each time step. This output provides an assessment of the group's situational awareness, including key indicators of Collaborative Problem-Solving (CPS) facets [25], levels of engagement, and a qualitative measure of the shared common ground. This enables the AI system to dynamically capture and model the group's evolving understanding, creating a cohesive and context-sensitive narrative of their interactions.

A typical example of a DNN in this context is the move classifier from [17]. This classifier is a multimodal, long short-term memory (LSTM)-based model designed to capture contextual information that conditions the sequence of cognitive states in a dialogue.

3.4 Workflow

To evaluate situational awareness in visual and audio data using the NSN, the workflow begins with the multimodal feature extractor, which processes the input data and returns features from various modalities based on task requirements, generating symbolic outputs for each modality. These symbolic outputs are then aligned and enriched through the MMDP mechanism, resulting in contextualized symbolic representations that integrate information across modalities. The MMDP output forms the foundation of the evolving CGS, which organizes information into evidence, facts, and questions to maintain shared understanding. Finally, the CGS is vectorized and passed into a recurrent DNN, which dynamically tracks the evolution of the common ground and generates outputs at each time step, providing situational awareness through indicators of group engagement, collaborative problem-solving, and qualitative assessments of the common ground.

4 Conclusion

This research builds on our prior work in understanding the complexities of multimodal interaction and addressing technical challenges in controlled environments. By fusing multimodal data streams into a coherent representation that evolves over time, our system aims to provide real-time support for more effective and natural collaboration in classrooms. This has the potential to transform learning experiences by fostering deeper understanding, personalized feedback, and more engaging collaborative activities.

5 Acknowledgements

This paper is based in part upon work supported by the National Science Foundation (NSF) under subcontracts to Colorado State University and Brandeis University on award DRL 2019805 (Institute for Student-AI Teaming), and by Other Transaction award HR00112490377 from the U.S. Defense Advanced Research Projects Agency (DARPA) Friction for Accountability in Conversational Transactions (FACT) program. Approved for public release, distribution unlimited. Views expressed herein do not reflect the policy or position of the National Science Foundation, the Department of Defense, or the U.S. Government.

References

1. Balwant, P.T.: The meaning of student engagement and disengagement in the classroom context: Lessons from organisational behaviour. *Journal of further and higher education* **42**(3), 389–401 (2018)
2. van Benthem, J., Fernández-Duque, D., Pacuit, E.: Evidence and plausibility in neighborhood structures. *Annals of Pure and Applied Logic* **165**(1), 106–133 (2014)

3. Bewersdorff, A., Hartmann, C., Hornberger, M., Sekler, K., Bannert, M., Kasneci, E., Kasneci, G., Zhai, X., Nerdel, C.: Taking the next step with generative artificial intelligence: The transformative role of multimodal large language models in science education. *Learning and Individual Differences* **118**, 102601 (2025)
4. Brdiczka, O., Langet, M., Maisonnasse, J., Crowley, J.L.: Detecting human behavior models from multimodal observation in a smart home. *IEEE Transactions on Automation Science and Engineering* **6**(4), 588–597 (2009). <https://doi.org/10.1109/TASE.2008.2004965>
5. Bredin, H., Yin, R., Coria, J.M., Gelly, G., Korshunov, P., Lavechin, M., Fustes, D., Titeux, H., Bouaziz, W., Gill, M.P.: Pyannote. audio: neural building blocks for speaker diarization. In: *ICASSP 2020-2020 IEEE International conference on acoustics, speech and signal processing (ICASSP)*. pp. 7124–7128. IEEE (2020)
6. Böck, R., Glüge, S., Wendemuth, A., Limbrecht, K., Walter, S., Hrabal, D., Traue, H.C.: Intraindividual and interindividual multimodal emotion analyses in human-machine-interaction. In: *2012 IEEE International Multi-Disciplinary Conference on Cognitive Methods in Situation Awareness and Decision Support*. pp. 59–64 (2012). <https://doi.org/10.1109/CogSIMA.2012.6188409>
7. Capallera, M., Angelini, L., Meteier, Q., Khaled, O.A., Mugellini, E.: Human-vehicle interaction to support driver’s situation awareness in automated vehicles: A systematic review. *IEEE Transactions on Intelligent Vehicles* **8**(3), 2551–2567 (2023). <https://doi.org/10.1109/TIV.2022.3200826>
8. Chen, J., Seng, K.P., Smith, J., Ang, L.M.: Situation awareness in ai-based technologies and multimodal systems: Architectures, challenges and applications. *IEEE Access* (2024)
9. D’Aniello, G., Gravina, R., Gaeta, M., Fortino, G.: Situation-aware sensor-based wearable computing systems: A reference architecture-driven review. *IEEE Sensors Journal* **22**(14), 13853–13863 (2022). <https://doi.org/10.1109/JSEN.2022.3180902>
10. Earl, S.R., Taylor, I.M., Meijen, C., Passfield, L.: Autonomy and competence frustration in young adolescent classrooms: Different associations with active and passive disengagement. *Learning and Instruction* **49**, 32–40 (2017)
11. Endsley, M.R.: A systematic review and meta-analysis of direct objective measures of situation awareness: A comparison of sagat and spam. *Human Factors* **63**(1), 124–150 (2021). <https://doi.org/10.1177/0018720819875376>, pMID: 31560575
12. Goel, S., Lymperopoulos, P., Thielstrom, R., Krause, E., Feeney, P., Lorang, P., Schneider, S., Wei, Y., Kildebeck, E., Goss, S., et al.: A neurosymbolic cognitive architecture framework for handling novelties in open worlds. *Artificial Intelligence* **331**, 104111 (2024)
13. Hegde, R.M., Kurniawan, J., Rao, B.D.: On the design and prototype implementation of a multimodal situation aware system. *IEEE Transactions on Multimedia* **11**(4), 645–657 (2009). <https://doi.org/10.1109/TMM.2009.2017631>
14. Joshi, G., Walambe, R., Kotecha, K.: A review on explainability in multimodal deep neural nets. *IEEE Access* **9**, 59800–59821 (2021). <https://doi.org/10.1109/ACCESS.2021.3070212>
15. Ju, U., Chuang, L.L., Wallraven, C.: Acoustic cues increase situational awareness in accident situations: A vr car-driving study. *IEEE Transactions on Intelligent Transportation Systems* **23**(4), 3281–3291 (2022). <https://doi.org/10.1109/TITS.2020.3035374>
16. Keshun, Y., Guangqi, Q., Yingkui, G.: Optimizing prior distribution parameters for probabilistic prediction of remaining useful life using deep learning. *Reliability Engineering & System Safety* **242**, 109793 (2024)

17. Khebour, I., Lai, K., Bradford, M., Zhu, Y., Brutti, R., Tam, C., Tu, J., Ibarra, B., Blanchard, N., Krishnaswamy, N., et al.: Common ground tracking in multimodal dialogue. arXiv preprint arXiv:2403.17284 (2024)
18. Krishnaswamy, N., Narayana, P., Bangar, R., Rim, K., Patil, D., McNeely-White, D., Ruiz, J., Draper, B., Beveridge, R., Pustejovsky, J.: Diana's world: A situated multimodal interactive agent. In: Proceedings of the AAAI Conference on Artificial Intelligence. vol. 34, pp. 13618–13619 (2020)
19. Lee, C.E., Baek, J., Son, J., Ha, Y.G.: Deep ai military staff: Cooperative battlefield situation awareness for commander's decision making. *The Journal of Supercomputing* **79**(6), 6040–6069 (2023)
20. Li, Q., Ng, K.K., Simon, C., Yiu, C.Y., Lyu, M.: Recognising situation awareness associated with different workloads using eeg and eye-tracking features in air traffic control tasks. *Knowledge-Based Systems* **260**, 110179 (2023)
21. Liu, S., Liu, H., John, V., Liu, Z., Blasch, E.: Enhanced situation awareness through cnn-based deep multimodal image fusion. *Optical Engineering* **59**(5), 053103–053103 (2020)
22. Lugaesi, C., Tang, J., Nash, H., McClanahan, C., Uboweja, E., Hays, M., Zhang, F., Chang, C.L., Yong, M.G., Lee, J., et al.: Mediapipe: A framework for building perception pipelines. arXiv preprint arXiv:1906.08172 (2019)
23. Mai, S., Hu, H., Xu, J., Xing, S.: Multi-fusion residual memory network for multimodal human sentiment comprehension. *IEEE Transactions on Affective Computing* **13**(1), 320–334 (2022). <https://doi.org/10.1109/TAFFC.2020.3000510>
24. Munir, A., Aved, A., Blasch, E.: Situational awareness: techniques, challenges, and prospects. *AI* **3**(1), 55–77 (2022)
25. Nelson, L.M.: Collaborative problem solving. In: *Instructional-design theories and models*, pp. 241–267. Routledge (2013)
26. Pacuit, E.: *Neighborhood semantics for modal logic*. Springer (2017)
27. Paul, P.P., Gavrilova, M., Klimenko, S.: Situation awareness through multimodal biometric template security in real-time environments. In: 2013 International Conference on Cyberworlds. pp. 82–88 (2013). <https://doi.org/10.1109/CW.2013.80>
28. Pustejovsky, J., Krishnaswamy, N.: Embodied human computer interaction. *KI-Künstliche Intelligenz* **35**(3), 307–327 (2021)
29. Pustejovsky, J., Krishnaswamy, N.: Multimodal semantics for affordances and actions. In: *International Conference on Human-Computer Interaction*. pp. 137–160. Springer (2022)
30. Pustejovsky, J., Zhu, Y.: Lexical event models for multimodal dialogues. In: *International Conference on Human-Computer Interaction*. pp. 174–192. Springer (2025)
31. Radford, A., Kim, J.W., Xu, T., Brockman, G., McLeavey, C., Sutskever, I.: Robust speech recognition via large-scale weak supervision. In: *International conference on machine learning*. pp. 28492–28518. PMLR (2023)
32. Rim, K., Tu, J., Ye, B., Verhagen, M., Holderness, E., Pustejovsky, J.: The coreference under transformation labeling dataset: entity tracking in procedural texts using event models. In: *Findings of the Association for Computational Linguistics: ACL 2023*. pp. 12448–12460 (2023)
33. Saad, A., Fouad, H., Mohamed, A.A.: Situation-aware recommendation system for personalized healthcare applications. *Journal of Ambient Intelligence and Humanized Computing* pp. 1–15 (2021)
34. Southwell, R., Pugh, S., Perkoff, M., Clevenger, C., Bush, J., Lieber, R., Ward, W., Foltz, P., D'Mello, S.: Challenges and feasibility of automatic speech recognition for

- modeling student collaborative discourse in classrooms. International Educational Data Mining Society (2022)
35. Tu, J., Rim, K., Ye, B., Lai, K., Pustejovsky, J.: Dense paraphrasing for multimodal dialogue interpretation. *Frontiers in artificial intelligence* **7**, 1479905 (2024)
 36. Yang, H., Wu, J., Hu, Z., Lv, C.: Real-time driver cognitive workload recognition: Attention-enabled learning with multimodal information fusion. *IEEE Transactions on Industrial Electronics* **71**(5), 4999–5009 (2024). <https://doi.org/10.1109/TIE.2023.3288182>
 37. Yang, J., Liang, N., Pitts, B.J., Prakah-Asante, K.O., Curry, R., Blommer, M., Swaminathan, R., Yu, D.: Multimodal sensing and computational intelligence for situation awareness classification in autonomous driving. *IEEE Transactions on Human-Machine Systems* **53**(2), 270–281 (2023). <https://doi.org/10.1109/THMS.2023.3234429>
 38. Zhu, Y., VanderHoeven, H., Lai, K., Bradford, M., Tam, C., Khebour, I., Brutti, R., Krishnaswamy, N., Pustejovsky, J.: Modeling theory of mind in multimodal hci. In: *International Conference on Human-Computer Interaction*. pp. 205–225. Springer (2024)